

# THE IMPERATIVE PARADIGM: A DECOMPOSITION OF VOICE MORPHOLOGY ACROSS FORMOSAN LANGUAGES\*

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This paper argues that “voice morphology” in Philippine-type voice systems should be attributed to multiple syntactic heads—Top, Mood, and Appl—and their various combinations, which are typically bundled together to create the appearance of complementary distribution. However, an examination of the imperative paradigm across Formosan languages reveals that voice forms are morphologically decomposable, and reflect with increasing complexity the structure underlying AV, PV/LV, and CV constructions. In particular, it will be argued that voice morphology at its core reflects grammatical mood distinctions—i.e. [REALIS] vs. [IRREALIS] features—in the syntax.

## 1. Introduction

In much of the literature on *Austronesian voice*—in particular “Philippine-type” voice systems—analyses tend to focus on syntactic phenomena such as case assignment and extraction patterns (see Rackowski 2002, Aldridge 2004, Chen 2017, Erlewine et al. 2017, a.o.), which has yielded invaluable insights and cross-linguistic comparisons. In these analyses, however, voice morphology is typically not given particular attention, and is often described as a set of morphemes in complementary distribution whose locus of spell-out is delegated to a single syntactic head.

This makes sense at first glance, given canonical examples of voice alternations (1).

- (1) *Voice alternations in indicative constructions (Tgdaya Seediq)*
- |    |                          |                |       |            |                           |
|----|--------------------------|----------------|-------|------------|---------------------------|
| a. | wada=[ku]                | <b>m</b> -ege  | patis | heya       |                           |
|    | PST=1SG.PIV              | AV-give        | book  | 3SG        |                           |
|    | ‘I gave him a/the book.’ |                |       |            | actor voice (AV)          |
| b. | wada=mu                  | biq- <b>un</b> | patis | [ka hey]   |                           |
|    | PST=1SG.GEN              | eat-PV         | book  | PIV 3SG    |                           |
|    | ‘I gave him a/the book.’ |                |       |            | patient voice (PV)        |
| c. | wada=mu                  | <b>s</b> -bege | heya  | [ka patis] |                           |
|    | PST=1SG.GEN              | CV-give        | 3SG   | PIV book   |                           |
|    | ‘I gave him the book.’   |                |       |            | circumstantial voice (CV) |

These examples display the classic function of voice systems, where verbal morphology (bolded) reflects which argument of the sentence—which can be core or non-core—is the *pivot* (brackets), a privileged argument in terms of case, word order, definiteness, extraction, and more. In Seediq, *actor voice* (AV) picks out agents and intransitive subjects, *patient voice* (PV) picks out transitive themes and ditransitive goals, while *circumstantial voice* (CV) picks out ditransitive themes and oblique arguments;<sup>1</sup> this correspondence is fairly consistent across languages, with some degree

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<sup>1</sup> An account of locative voice (LV) will largely be set aside in this paper; see section 5 for discussion of reasons.

of variation. Overall, it does in fact appear that voice morphology is in complementary distribution, where each morpheme is distinct in form and serves a complementary function.

Despite these apparent facts, this paper takes a morphological approach to voice systems and argues that “voice morphology” should be attributed to multiple distinct syntactic heads and their various combinations. In particular, I offer evidence based on the comparison of indicative and imperative voice paradigms in Formosan languages (2), focusing on the latter. I argue that many of these forms are clearly morphologically decomposable, and that a sort of containment relation holds across the voices with increasing complexity from AV to PV/LV to CV. Assuming a morphology-to-syntax mapping under the Mirror Principle (Baker 1985)—where morphological complexity reflects syntactic complexity—this suggests that voice forms involve varying degrees of structure, parallel to the morphology, with increasing complexity from AV to PV/LV to CV.

I propose that voice involves the interaction of three functional projections: Top, Mood, and Appl, which are systematically present in some but not all voice constructions. In particular, AV has Mood, PV has Top and Mood, and CV has all three Top, Mood, and Appl projected. The difference between the indicative and imperative voice paradigms, therefore, is merely a surface morphological difference: any structural complexity is obscured in the former because the relevant heads are bundled together during spell-out, but becomes apparent in the latter when these heads are decomposed and spelled-out individually. This divergence is mediated by the features of the Mood head—i.e. [REALIS] vs. [IRREALIS]—which forms the basis of all voice morphology.

(2) *Imperative voice paradigm in Formosan languages (simplified)*<sup>2,3</sup>

|        | INDICATIVE |           |           |           | IMPERATIVE |           |           |           |
|--------|------------|-----------|-----------|-----------|------------|-----------|-----------|-----------|
|        | <u>AV</u>  | <u>PV</u> | <u>LV</u> | <u>CV</u> | <u>AV</u>  | <u>PV</u> | <u>LV</u> | <u>CV</u> |
| Seediq | <m>        | -un       | -an       | s-        | ∅          | -i        | -i        | -ani      |
| Bunun  | ma-        | -un       | -an       | 'is-      | -a         | -av       | -av       | 'is- -av  |
| Puyuma | <em>       | -aw       | -ay       | -anay     | ∅          | -u        | -i        | -an       |
| Tsou   | <m>        | -a        | -i        | -neni     | -          | -         | -         | -         |
| Paiwan | <em>       | -in       | -an       | si-       | -u         | -u        | -i        | -an       |
| Amis   | <om>       | -en       | -an       | sa-       | ∅          | -en       | -         | sa- -en   |

This analysis draws in many ways from existing insights on the nature of voice systems, including work on morpheme decomposition in Malagasy (Travis 1991, 2010, Pearson 2005) as well as applicative analyses of CV constructions (Rackowski 2002, Aldridge 2004, Rackowski & Richards 2005, Chang 2009), and topic-like characterization of pivots (Schachter 1976, Pearson 2005, Chen 2017) in Austronesian more broadly. Thus, the fundamental parts of this analysis are not novel by any means, but the contribution here is to bring these insights together to articulate in more precise terms the functional projections responsible for voice systems and their complex alternations, such that the morphology and syntax are consistent with one another. This offers as well a potential compromise between argument structural and information structural approaches to voice systems, since it incorporates elements from both sides of the debate.

<sup>2</sup> The voice forms included in this paradigm are simplified for exposition—see the sources below for discussion of allomorphy and other complexities. However, this should not affect the generalizations put forth in this paper.

<sup>3</sup> Sources: 2018 CIP grammar series, 2024 *Handbook on Formosan Languages*, eds. Li, Zeitoun, & de Busse

Furthermore, this paper pursues an existing observation that voice morphology at its core reflects *grammatical mood* (cf. Ross 2009, Aldridge 2015, 2021). While the realis/irrealis mood distinction has long been noted within Austronesian historical literature, it has yet been formalized as part of the synchronic derivation of voice systems. This paper takes on such an analysis, where voice represents underlying mood alternations, which can then be bundled with additional mechanisms such as *topicalization* and *applicativization*. Since this is a preliminary proposal and focuses on morphological decomposition, further work is required to connect this account back to established syntactic observations about voice systems, such as case and extraction patterns.

It is important to note that beyond the data presented here, there are various other reasons that motivate the decomposition of voice morphology: the different orientation of voice markers (prefix vs. suffix), the placement of voice projections in the clausal hierarchy based on affix and auxiliary order (Wu 2024), the distribution of voice markers in complementation contexts (Chen 2010, Chang 2017), and lastly their distinct historical origins (Starosta et al. 1981, Beguš 2016). These topics warrant their own discussion, but the key point here is that paying closer attention to morphology can offer insights to complex syntactic phenomena, and get us further in relating Austronesian voice systems to the basic building blocks of grammar across languages.

In the remainder of this paper, section 2 will introduce data on indicative and imperative voice paradigms, section 3 will discuss theoretical assumptions on syntactic bundling, section 4 will derive the different voice alternations, and section 5 will conclude and address next steps.

## 2. The Imperative Paradigm

In Formosan languages, verbs have distinct sets of voice forms for *indicative* (1) and *imperative* (3) constructions. The full range of voice alternations is available in imperative contexts—while the addressee is fixed as the (implicit) agent, the pivot can target any argument of the sentence.

### (3) Imperative constructions have irrealis voice forms (Tgdaya Seediq)

- |               |      |                 |           |                     |                  |
|---------------|------|-----------------|-----------|---------------------|------------------|
| a. Ø-ekan     | ido! | b. puq-i        | [ka ido]! | c. biq-an-i         | yaku [ka patis]! |
| AV.IRR-eat    | rice | eat-PV.IRR      | PIV rice  | give.CV.IRR         | 1SG PIV book     |
| ‘Eat (rice)!’ |      | ‘Eat the rice!’ |           | ‘Give me the book!’ |                  |

This paper will draw primarily from data on Seediq, where the voice forms in imperative constructions are the same as those in *negative* (4) and *prohibitive* (i.e. negative imperative) (5) constructions (Holmer 1996). This is labelled as an irrealis (IRR) form, to encompass actions that are untrue or unknown because they either did not or have not yet happened (Bybee 1998).

### (4) Negative constructions have irrealis voice forms (Tgdaya Seediq)

- |                        |            |      |                           |            |            |
|------------------------|------------|------|---------------------------|------------|------------|
| a. ini=[ku]            | Ø-ekan     | ido  | b. ini=mu                 | puq-i      | [ka hlama] |
| NEG=1SG.PIV            | AV.IRR-eat | rice | NEG=1SG.GEN               | eat-PV.IRR | PIV candy  |
| ‘I didn’t eat (rice).’ |            |      | ‘I didn’t eat the candy.’ |            |            |

### (5) Prohibitive constructions have irrealis voice forms (Tgdaya Seediq)

- |              |            |                        |            |            |
|--------------|------------|------------------------|------------|------------|
| a. iya       | Ø-ekan!    | b. iya                 | puq-i      | [ka hlama] |
| NEG.IMP      | AV.IRR-eat | NEG.IMP                | eat-PV.IRR | PIV candy  |
| ‘Don’t eat!’ |            | ‘Don’t eat the candy!’ |            |            |

The indicative and imperatives voice forms do not appear to be morphologically related, and there has been literature explaining the diachronic development of these separate paradigms (Starosta et al. 1981, Ross 2009, Aldridge 2015, 2021). Interestingly, if we took the imperative paradigm to be our starting point, we would not immediately assume that the voice forms are in complementary distribution. These paradigms, as seen in (2), are reprinted below (6) and altered to reflect a morphologically decomposed parsing, where some forms are internally complex and contain sub-parts that are found in other forms. In particular, there appears to be a containment relation of sorts between the voice forms, where it gets increasingly complex and builds on itself from AV to PV/LV to CV. This is seen in Seediq, Bunun, Puyuma, and Amis: the AV form is either null or a single morpheme, while the PV/LV form introduces an additional morpheme, and the CV form adds yet another morpheme on top of that. For example, in Seediq, the AV.IRR form is  $\emptyset$ , the PV.IRR/LV.IRR form is *-i*, and the CV.IRR form is *-an-i*. A similar pattern can also be observed in the indicative paradigms of Puyuma and Tsou, though it does not fully extend to AV forms.

(6) *Imperative voice paradigm in Formosan languages, decomposed (simplified)<sup>2,3</sup>*

|        | INDICATIVE    |             |             |                | IMPERATIVE  |             |             |                  |
|--------|---------------|-------------|-------------|----------------|-------------|-------------|-------------|------------------|
|        | <u>AV</u>     | <u>PV</u>   | <u>LV</u>   | <u>CV</u>      | <u>AV</u>   | <u>PV</u>   | <u>LV</u>   | <u>CV</u>        |
| Seediq | < <i>m</i> >  | <i>-un</i>  | <i>-an</i>  | <i>s-</i>      | $\emptyset$ | <i>-i</i>   | <i>-i</i>   | <i>-an-i</i>     |
| Bunun  | <i>ma-</i>    | <i>-un</i>  | <i>-an</i>  | <i>'is-</i>    | <i>-a</i>   | <i>-a-v</i> | <i>-a-v</i> | <i>'is- -a-v</i> |
| Puyuma | < <i>em</i> > | <i>-a-u</i> | <i>-a-i</i> | <i>-an-a-i</i> | $\emptyset$ | <i>-u</i>   | <i>-i</i>   | <i>-an-i</i>     |
| Tsou   | < <i>m</i> >  | <i>-a</i>   | <i>-i</i>   | <i>-nen-i</i>  | -           | -           | -           | -                |
| Paiwan | < <i>em</i> > | <i>-in</i>  | <i>-an</i>  | <i>si-</i>     | <i>-u</i>   | <i>-u</i>   | <i>-i</i>   | <i>-an</i>       |
| Amis   | < <i>om</i> > | <i>-en</i>  | <i>-an</i>  | <i>sa-</i>     | $\emptyset$ | <i>-en</i>  | -           | <i>sa- -en</i>   |

A potential question to ask is whether there is an identifiable imperative marker here—this will later on inform the claim that there is something inherent to voice that reflects mood distinctions.

The crucial observation here, is that voice forms can involve up to three distinct parts, as seen above for CV forms in Bunun and Puyuma. Furthermore, Seediq also displays three distinct parts elsewhere in its grammar: in *hortative* constructions (7), an extra *-a* morpheme is added to the regular irrealis voice forms, making them more specialized irrealis forms in a sense. The AV form is *-a*, the PV/LV form is *-a-i*, and the CV form is *-an-a-i*, which bears striking parallels to the Puyuma indicative forms. Note that this is partially obscured in the Tgdaya dialect of Seediq due to the phonological coalescence of /-a-i/ as [-e] (8), but is seen distinctly in the Truku dialect (9).

(7) *Hortative constructions have irrealis voice forms (Tgdaya Seediq)*

- a. sa-**a**=ta!  
go-AV.HORT=1PL.INCL  
'Let's go!' AV.HORT *-a* < HORT *-a* + AV.IRR  $\emptyset$
- b. sa-**e**=ta [ka Holi]  
go-PV.HORT=1SG.INCL PIV Puli  
'Let's go to Puli.' PV.HORT *-e* < HORT *-a* + PV.IRR *-i*
- c. biq-**an-e**=mu isu [ka patis]  
give-CV.HORT=1SG.GEN 2SG PIV book  
'Let me give you the book.' CV.HORT *-ane* < HORT *-a* + CV.IRR *-an-i*

(8) *Voice markers in realis and irrealis moods (Tgdaya Seediq)*

|                      | <u>AV</u>    | <u>PV</u>   | <u>LV</u>   | <u>CV</u>     |
|----------------------|--------------|-------------|-------------|---------------|
| REALIS               | < <i>m</i> > | - <i>un</i> | - <i>an</i> | <i>s-</i>     |
| IRREALIS (DEFAULT)   | ∅            | - <i>i</i>  | - <i>i</i>  | - <i>an-i</i> |
| IRREALIS (HORTATIVE) | - <i>a</i>   | - <i>e</i>  | - <i>e</i>  | - <i>an-e</i> |

(9) *Voice markers in realis and irrealis moods (Truku Seediq) (Tsukida 2009, adapted)*

|                      | <u>AV</u>     | <u>PV</u>    | <u>LV</u>    | <u>CV</u>       |
|----------------------|---------------|--------------|--------------|-----------------|
| REALIS               | < <i>em</i> > | - <i>un</i>  | - <i>an</i>  | <i>se-</i>      |
| IRREALIS (DEFAULT)   | ∅             | - <i>i</i>   | - <i>i</i>   | - <i>an-i</i>   |
| IRREALIS (HORTATIVE) | - <i>a</i>    | - <i>a-i</i> | - <i>a-i</i> | - <i>an-a-i</i> |

Interestingly, the *-a* morpheme shows up repeatedly throughout Formosan languages, such as in Bunun, Puyuma, and Tsou, and has an interesting placement between the other morphemes in the CV.HORT form *-an-a-i*. This can be reconstructed in Proto-Austronesian as the non-finite marker *\*-a* (Aldridge 2021), and will be relevant later on in regards to mood and clause structure.

It should hopefully be clear that voice forms can be morphologically complex, often seen in the imperative paradigm but in the indicative paradigm of some languages as well. The next logical question, then, is what these individual parts represent in the syntax, and how to simultaneously account for the contrast in morphological behavior: sets of voice forms that are in complementary distribution, versus those that show internal decomposition. In the next two sections, I address these issues through the bundling of functional heads in varying base structures.

### 3. Theoretical Assumptions

The key theoretical assumption in this paper is that functional projections can be *bundled* or fully articulated depending on the language and/or construction, which impacts surface morphological outcomes. This idea has been applied to many different areas of syntax, including the argument structural domain (“Voice-bundling”; Pyllkänen 2008, Harley 2017) and the agreement domain (“Split-IP”; Thráinsson 1996, Bobaljik & Thráinsson 1998), among others. This section explores the application of this idea towards complex structures found in imperatives, and the next section will extend this idea towards complex structures found in Austronesian voice.

#### 3.1. Imperatives

The current understanding of imperatives is that they are syntactially and semantically complex (see Han 1999, Rivero and Terzi 1995, Kaufmann 2012, Zanuttini et al. 2012, a.o.), and consist of three main components: 1) a speech act  $C_{[IMP]}$  head, 2) a necessity modal  $Mod$  head, and 3) an irrealis mood  $Mood_{[IRR]}$  head. In particular, recent work has shown that while these components are often bundled—i.e. languages commonly have dedicated imperative verbal forms—it is also possible for these components to come apart and surface in other morphological forms.

This can be seen in the contrast between imperatives and prohibitives in some languages, namely in the subjunctive repair strategies of imperatives in negative contexts (Fenger 2025). In Spanish, for example, while the positive imperative (10a) has regular imperative morphology and

precedes the pronominal clitics =*me* and =*lo*, the negative imperative (10b) has subjunctive morphology and is instead ordered after the negative particle *no* and the clitics *me* and *lo*.

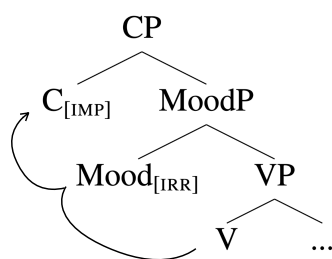
(10) *Imperatives have different verb forms in positive and negative contexts (Spanish)*

- |    |                  |      |      |    |                        |     |     |               |
|----|------------------|------|------|----|------------------------|-----|-----|---------------|
| a. | dá               | =me  | =lo! | b. | no                     | me  | lo  | des!          |
|    | give.IMP.2SG     | =1SG | =3SG |    | NEG                    | 1SG | 3SG | give.SUBJ.2SG |
|    | ‘Give it to me!’ |      |      |    | ‘Don’t give it to me!’ |     |     |               |

In addition to asking why positive and negative imperative verbs differ morphologically, another interesting question here is why the latter employs an entirely different form, which is also used elsewhere in non-imperative contexts in the language—a potential answer is one of bundling. In regular imperative structures (11a), the verb moves through all three projections (only  $C_{[IMP]}$  and  $Mood_{[IRR]}$  shown below), which are completely bundled and spelled-out altogether as “imperative morphology”. In negative imperative structures (11b), the projection of the negative particle *no* interrupts head movement and subsequent bundling, such that the verb moves to  $Mood_{[IRR]}$  and is instead spelled-out as “subjunctive morphology”. Note that the difference in head movement is corroborated by the relative order of the verb with respect to the pronominal clitics. Overall, this means that subjunctive morphology is actually a reflex of the irrealis  $Mood_{[IRR]}$  head, which thus explains its wider distribution in irrealis environments outside of imperative constructions.

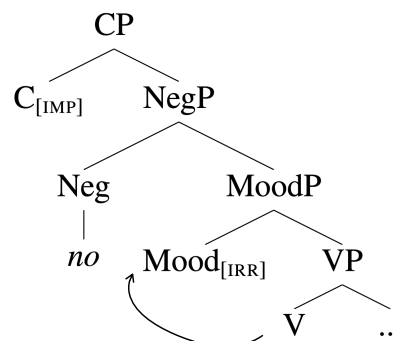
(11) *Structure of positive vs. negative imperative (Spanish)*

a. positive imperative



$V + Mood_{[IRR]} + C_{[IMP]} \rightarrow$  “imperative”

b. negative imperative



$V + Mood_{[IRR]} \rightarrow$  “subjunctive”

The crucial insight here, therefore, is that two surface forms may look unrelated but have overlap in their underlying structures, in the configuration where one spells-out a “subset” of the structure of another. This relies on the idea that vocabulary items can target individual or bundled heads in the syntax for spell-out, such as through *portmanteaus* or *spans* (Svenonius 2016). This has two relevant consequences: that morphology can be underspecified with respect to syntactic structure if there are no (overt) vocabulary items to spell-out certain heads, and that morphology can obscure complexities in the syntax if vocabulary items bundle multiple heads at once.

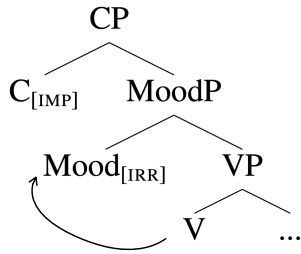
### 3.2. Irrealis Mood in Formosan

That being said, we would expect languages that do not employ bundling in imperative forms to not have distinct morphological strategies for positive and negative imperatives. Coming back to

Formosan languages, we can straightforwardly account for the irrealis voice forms that are shared across imperative, negative, and prohibitive constructions as a case of *underspecification*. This is schematized below (12), where the constructions all share a common Mood<sub>[IRR]</sub> projection despite differences in the larger structure, such as the Neg or C<sub>[IMP]</sub> projections. Since there are no (overt) vocabulary items for C<sub>[IMP]</sub> and no bundled vocabulary items for Mood<sub>[IRR]</sub> and C<sub>[IMP]</sub>, this leads to the structures being spelled-out uniformly with “irrealis morphology”. In other words, the irrealis forms in Seediq are a direct parallel to subjunctive repair strategies in other languages, where the morphology only spells-out a subset of the structure and is thus systematically underspecified.

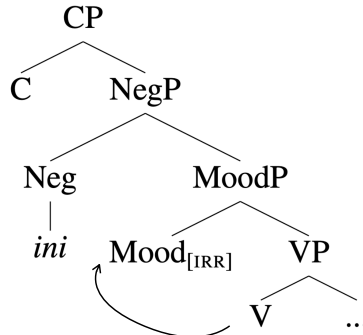
(12) *Structure of different constructions in irrealis mood (Tgdaya Seediq)*

a. imperative



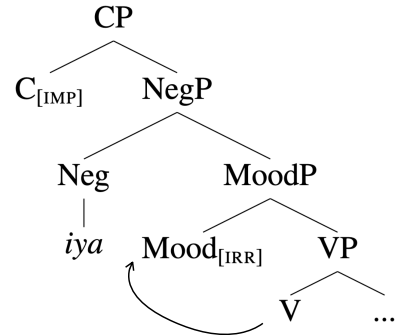
V+Mood<sub>[IRR]</sub> → “irrealis”

b. negative



V+Mood<sub>[IRR]</sub> → “irrealis”

c. prohibitive



V+Mood<sub>[IRR]</sub> → “irrealis”

However, while Seediq does not do bundling for the purposes of imperative morphology, we will see in the next section that it does so in the case of voice morphology. It is important that we have established the source of irrealis forms here as the Mood<sub>[IRR]</sub> head, as it will be relevant when we explore the contrast in voice morphology between the indicative and imperative voice paradigms. The fact that these paradigms are *suppletive*—i.e. do not appear morphologically related—suggests that grammatical mood underlies a critical component of voice morphology.

#### 4. Proposed Analysis: Bundling vs. Decomposition

This section explores the analysis that “voice” is underlyingly complex—in a fashion similar to imperatives—and can be decomposed into three main functional heads: Top, Mood, and Appl. The difference between the indicative and imperative voice paradigms is one of bundling versus decomposition, which is triggered by features present on the Mood head: [REAL] vs. [IRR]. One critical aspect to this analysis is that not all of these heads are projected across the different voice structures, with systematic differences in complexity between AV, PV/LV, and CV.

##### 4.1. Top, Mood, Appl

I assume here a morphology-to-syntax mapping under the Mirror Principle (Baker 1985), where morphological complexity reflects structural complexity. Therefore, given that voice morphology can involve up to three distinct morphemes (as seen in section 2), this suggests that the structure of voice can involve up to three distinct projections. The proposal here is that voice phenomena are syntactically and semantically complex, and can be decomposed into three main components:

1) a middle topic Top head, 2) a mood Mood head, and 3) an applicative Appl head.

The relative order of these projections within the clausal hierarchy is shown in (13). This is inspired by work on morpheme decomposition in Malagasy (Travis 1991, 2010, Pearson 2005) which uses Asp, E, and Appl as voice heads; the specific motivations and analysis in the present paper departs from the original work, but the decomposition and relative order remain the same.

(13) *Hierarchy of Projections*

C > T > Neg > Asp > **Top** > **Mood** > Voice > **Appl** > v > √ROOT

The basis for these projections is not arbitrary or new, but draws on well-known insights in the literature about voice systems. First, the Top head is responsible for pivot promotion, and reflects the fact that pivots are topic-like elements (Schachter 1976, Pearson 2005, Chen 2017). Note that this specific topic projection is not located in the *left-periphery* but in the *middle field*, which is a phenomenon that has been described in various other languages (Paul 2002, LaCerta 2020). The motivations for this is not the focus of the paper, but is based on evidence from affix and auxiliary order (Wu 2024) and is akin to object shift accounts of voice where pivots move to the edge of the verbal phase (Rackowski & Richards 2005). Second, the Mood head hosts realis and irrealis features, a distinction that has been reconstructed to Proto-Austronesian (Ross 2009, Aldridge 2015). This distinction has also be conceived as one of non-finiteness (Aldridge 2021, cf. Starosta et al. 1981, Travis 2010), and there is evidence from voice marking in complementation and reduced clauses that would support a similar approach (Chen 2010, Wurmbrand 2016). Third, the Appl head is a common analysis of CV constructions (Rackowski 2000, Aldridge 2004, Chang 2009), and is responsible for introducing oblique arguments and reordering core internal arguments, two functions that have been attributed to applicatives cross-linguistically.<sup>4</sup>

Overall, there are interesting discussions beyond the scope of this paper about the middle field placement of discourse-related functional projections, such as those for topic and grammatical mood. This relates to cross-linguistic phenomena such as object shift, scrambling, and more (cf. “low-IP area”, Belletti 2004; “*mittelfeld*”, Frey 2004), and bears on the syntax of movement and phases. Nevertheless, the main takeaway in this paper is not the nature or placement of these particular projections, but rather the need for decomposition and the relative order of these heads.

#### 4.2. Voice derivation

Voice alternations can thus be captured by structural differences in whether the three components—Top, Mood, and Appl—are projected, with increasing complexity from AV to PV/LV to CV. The structure remains unchanged for each voice construction between the indicative and imperative paradigms, and instead the crucial distinction is in the features of the Mood head—i.e. [REALIS] vs. [IRREALIS]—that affect which vocabulary items are available for insertion and bundling.

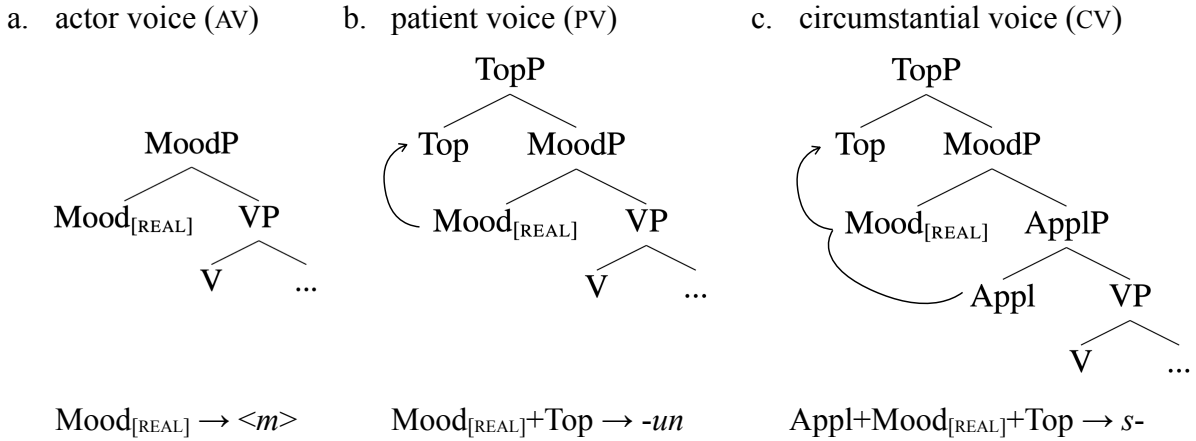
The indicative case is illustrated below (14), showing the difference in structure between AV, PV, and CV. The Mood head is specified with a [REALIS] feature. In AV constructions, only the Mood<sub>[REAL]</sub> head is projected, which is spelled-out straightforwardly as the AV infix <*m*>; in other words, basic forms of indicative AV morphology can be thought of as realis mood markers. The

<sup>4</sup> I remain agnostic about the characterization of Appl—in the literature there are two main analyses: 1) an argument introducing applicative (Pylkkänen 2000, Rackowski 2002, Aldridge 2004), and 2) a raising applicative (Georgala 2012, Nie 2019). This requires discussion of binding and argument structure which is not the focus of the paper, but the insights here about decomposition should still hold. See section 4.2 for some more discussion of these issues.



pivots in these constructions are either agents or intransitive subjects, and the assumption here is that the Top head is not projected, based on the reasoning that it does not create any interpretational difference (cf. Travis 1984 on Germanic V2, where subjects raise to Spec,TP but objects raise to Spec,CP). This means that AV constructions are a kind of *default* (cf. Wurmbrand 2016), which fits well with the observation that a binary distinction exists between AV and non-AV forms in independent parts of the grammar, such as in adverbs and auxiliaries (Chang 2009).

(14) *Structure of voice alternations in realis mood (Tgdaya Seediq)*



In PV constructions, both Mood and Top heads are projected, which are bundled together and spelled-out as the PV suffix *-un*; in other words, indicative PV morphology represents a combination of realis mood and topicalization. The pivots in these constructions are either transitive themes or ditransitive goals, which must be moved to Spec,TopP to receive topical interpretation. This means that PV constructions are the most basic case of non-AV constructions, promoting the default highest internal argument in transitives and ditransitives. In CV constructions, by contrast, all three Mood, Top, and Appl heads are projected, which are bundled and spelled-out altogether as the CV prefix *s-*; in other words, indicative CV morphology packages information about realis mood, topicalization, and applicativization all at once. The pivot in these constructions is either a ditransitive theme or oblique argument, which is made possible because Appl has the function of reordering internal arguments or introducing oblique arguments, such that it changes what the highest internal argument is that moves into Spec, TopP. This means that CV constructions are a specific subcase of non-AV constructions (cf. Chang 2009, Bogren Svensson 2023).

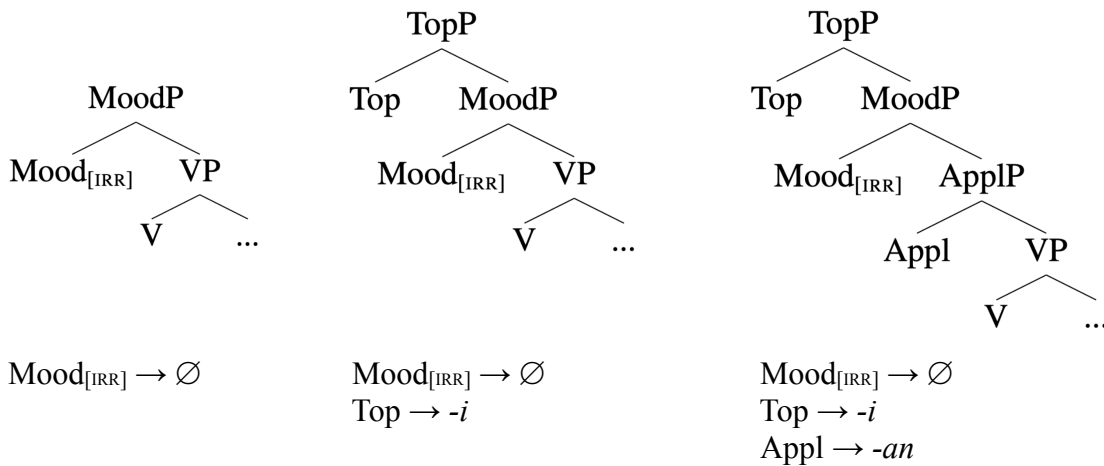
Despite the differences in structural complexity, we see that it is not apparent in the morphology of the indicative paradigm because of bundling. The realis vocabulary items AV <*m*>, PV *-un*, and CV *s-* spell-out increasingly greater portmanteuxs or spans, which obscure the complex structures. This gives voice morphology the appearance of complementary distribution, which is why they have commonly been ascribed to a single syntactic head in the literature. However, the complex structures become apparent in the imperative paradigm when bundling is unavailable.

The imperative cases are illustrated below (15), where the main structures for AV, PV, and CV remain unchanged, but the Mood head is now specified with an [IRREALIS] feature. Therefore, the vocabulary items that bundled with realis mood no longer apply, and a different set of items are inserted instead. In AV constructions, the Mood<sub>[IRR]</sub> head is projected, which is spelled-out as a null morpheme  $\emptyset$ ; in other words, basic imperative AV morphology simply reflects irrealis mood. In PV constructions, both Mood and Top heads are projected, which are spelled-out individually

as the null irrealis morpheme  $\emptyset$  and the Top suffix *-i*, respectively. In CV constructions, all three Mood, Top, and Appl heads are projected, which are spelled-out individually as the null irrealis morpheme  $\emptyset$ , the Top suffix *-i*, and the Appl suffix *-an*, respectively. Finally, recall the hortative voice forms from earlier, which were regular irrealis forms with an extra hortative suffix *-a*. This can easily be captured by positing a Mood<sub>[HORT]</sub> head that also combines with Top and Appl in the same fashion as Mood<sub>[IRR]</sub>, thus prompting the same morphologically decomposed behavior. One crucial point is that these three projections and their relative structural height predicts the correct affix order, in terms of which morphemes are closest to the verb root: Appl *-an* < Mood *-a* < Top *-i*. Overall, in the imperative (and hortative) paradigms, we see that morphological and syntactic complexity are more aligned, and reflect the presence of three distinct functional projections.

(16) *Structure of voice alternations in irrealis mood (Tgdaya Seedq)*

- a. actor voice (AV)      b. patient voice (PV)      c. circumstantial voice (CV)



The complete list of vocabulary items for voice morphology in Seediq are shown below, both for the realis (16a) and irrealis (16b) paradigms. A crucial observation to be made is that AV, PV, and CV are not morphological or syntactic primitives, but are made up of successively complex structures and vocabulary items; AV is contained within PV, which is contained within CV. Instead, the basic pieces of structure can be attributed to three functional projections, Top, Mood, and Appl, whose different combinations derive the voice alternations. In the realis forms they are bundled together, if present, and in the irrealis forms they are not. During vocabulary insertion, the vocabulary items are ordered such that more specific bundles are chosen over less specific ones, and Top, Mood, and Appl are only spelled-out independently when no bundle exists.

(16) *Vocabulary items (Tgdaya Seediq)*

- a. Mood<sub>[REAL]</sub> ↔ <*m*>      b. Mood<sub>[IRR]</sub> ↔  $\emptyset$       (Mood<sub>[HORT]</sub> ↔ *-a*)  
 Mood<sub>[REAL]</sub>+Top ↔ *-un*      Top ↔ *-i*  
 Appl+Mood<sub>[REAL]</sub>+Top ↔ *s-*      Appl ↔ *-an*

Finally, an interesting consequence of this analysis is that voice morphology represents underlying *grammatical mood* features, which can be bundled with additional mechanisms such as *topicalization* and *applicativization*. This builds on existing insights in the literature but brings them together in a novel way, and suggests that many theories of Austronesian voice are each on

the right track in their own ways, and we just have to find a compromise between them.

#### 4.3. Other Formosan Languages

The account developed above extends readily to other Formosan languages, which illustrates the range of surface morphological realizations that can be captured by such an underlying structure. The vocabulary items for Bunun (17), Puyuma (18), and Paiwan (19) are given below.

(17) *Vocabulary items (Bunun)*

- |                                                                                                                                      |                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| a. Mood <sub>[REAL]</sub> ↔ <i>ma-</i><br>Mood <sub>[REAL]</sub> +Top ↔ <i>-un</i><br>Appl+Mood <sub>[REAL]</sub> +Top ↔ <i>'is-</i> | b. Mood <sub>[IRR]</sub> ↔ <i>-a</i><br>Top ↔ <i>-v</i><br>Appl ↔ <i>'is-</i> |
|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|

(18) *Vocabulary items (Puyuma)*

- |                                                                                              |                                                                                      |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| a. Mood <sub>[REAL]</sub> ↔ <i>&lt;em&gt;</i><br>Mood <sub>[REAL]</sub> ↔ <i>-a / __</i> Top | b. Mood <sub>[IRR]</sub> ↔ $\emptyset$<br>Top ↔ <i>-u or -i</i><br>Appl ↔ <i>-an</i> |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|

(19) *Vocabulary items (Paiwan)*

- |                                                                                                                                            |                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| a. Mood <sub>[REAL]</sub> ↔ <i>&lt;em&gt;</i><br>Mood <sub>[REAL]</sub> +Top ↔ <i>-in</i><br>Appl+Mood <sub>[REAL]</sub> +Top ↔ <i>si-</i> | b. Mood <sub>[IRR]</sub> ↔ <i>-u</i><br>Mood <sub>[IRR]</sub> +Top ↔ <i>-i</i><br>Appl+Mood <sub>[IRR]</sub> +Top ↔ <i>-an</i> |
|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|

First, Bunun displays a pattern similar to Seediq, with complementary distribution in the indicative paradigm and decomposed morphology in the imperative paradigm. Second, Puyuma has an interesting case of decomposition in both paradigms, where, aside from the indicative AV infix *<em>*, the forms are completely derivable from each other and differ only in the presence of an extra *-a* in the realis paradigm. As noted earlier, this *-a* shows up in different places throughout Formosan languages to mark mood distinctions (Teng 2008, Aldridge 2015), such as in Tsou, Bunun, and Seediq, even if the exact function varies somewhat. The Puyuma paradigms can thus be captured by positing two different Mood<sub>[REAL]</sub> allomorphs, a default AV infix *<em>* and a suffix *-a* in non-AV contexts, while the rest of the vocabulary items are similar to the irrealis in Seediq. Third, Paiwan has the opposite case with bundling in both paradigms, where both Mood<sub>[REAL]</sub> and Mood<sub>[IRR]</sub> form portmanteaus/spans with Top and Appl, and there is no decomposition.

Similar analyses can be extended to the rest of the Formosan languages, most of which display some level of decomposition akin to Seediq, suggesting that the structures proposed here are fairly consistent cross-linguistically. Note that there are syncretisms in these paradigms that have not yet been addressed and seem more irregular, which will be left for future work.

#### 5. Locative Voice (LV)

One omission throughout the analysis in this paper is an account of *locative voice* (LV), which is slightly complicated for several reasons. On the one hand, it often has the same level of morphological complexity as PV; for instance, PV and LV both have the same imperative forms in Seediq and Bunun, or differ by only a vowel in most of the indicative and imperative paradigms. On the other hand, LV is often analyzed as having the same structural complexity as CV, where the same

applicative analysis holds and Appl introduces locative arguments in the case of LV. The trouble is thus how to reconcile such a difference. One possibility is that Appl can have specific features such as [LOC] and [OBL] that can give rise to different spell-outs and instances of bundling for LV and CV. In Seediq, then, this can be captured by having Appl<sub>[LOC]</sub> be part of another bundle in the indicative paradigm, but not having an overt morpheme in the imperative paradigm. The details of this need to be worked out further and will not be pursued in this paper, but it is unclear if this approach can capture the generalization that PV and LV tend to be closer morphologically.

Another complication to mention is that the distinction between PV and LV has shifted to an aspectual one in Seediq (Tsukida 2005), and LV is no longer productive in the canonical sense as a locative. This distinction is illustrated below (20), where PV has a perfective interpretation and LV has an imperfective one. One possibility is that PV and LV involve bundling with an Asp head instead, and that LV does not involve applicative structures at all. This would explain morphological levelling in the imperative paradigm between PV and LV, as irrealis forms typically do not make aspectual distinctions. However, this analysis would not extend to languages where LV constructions are productive in the traditional sense of promoting locative arguments as pivots. Thus, it appears that there is significant variation in the form and function of LV constructions across Formosan languages, which may be an interesting avenue to pursue in future research.

(10) *Aspectual distinctions between PV and LV constructions (Tgdaya Seediq) (Holmer 1996)*

- |                                                                                                                |                                                                                                           |
|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| a. wada=mu    mah- <b>un</b> [biru nii]<br>PST=1SG.GEN drink-PV beer this<br>'I have been drinking this beer.' | b. wada=mu    mah- <b>an</b> [biru nii]<br>PST=1SG.GEN drink-LV beer this<br>'I have drunk up this beer.' |
|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|

Overall, the contribution of this paper is to argue for a need for functional decomposition in the structure of voice, but more work is needed to determine exactly how this interacts with the syntax of argument structure and pivot selection in each voice construction. In particular, the structures proposed here must be able to capture the range of pivot selections in (in)transitives, ditransitives, causatives, control structures, and more. Another possibility, on the flip side, is that understanding these voice structures might in turn give us insights into argument structure.

## **6. Conclusion**

Overall, this paper takes a morphological approach to Austronesian voice, in a way that reveals potential insights about the syntax of voice systems. In particular, the position advocated here is that “voice” is a complex phenomenon that can be attributed to multiple functional heads in the structure: Top, Mood, and Appl. Across the voice alternations—from AV to PV/LV to CV—there is increasing morphological complexity, which can be captured by increasing structural complexity in terms of which of the three heads above are projected in each construction. Furthermore, these heads can be more or less transparent depending on the availability of bundling during spell-out. The one head that is systematically present is Mood, which means that voice morphology is at its core a reflection of realis or irrealis mood, with additional topicalization or applicativization. An important contribution of this work is the attempt to relate the “unusual” phenomenon of voice to well-known building blocks across languages, even if packaged in novel or unexpected ways.

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